



CS 329Z: Engineering AI Agents

Stanford / Fall 2026

Course Project

The final project is the main assignment of the course. Working in teams, you will design, build, and evaluate an agentic system of your own choosing over the quarter. Projects must relate in a reasonable way to at least one of the central topics of the course: building agents (retrieval, tool use, memory, multi-agent systems, optimization), the data agents need, or evaluating and making agents safe. Beyond that, the choice of domain is yours. Example project ideas:

- A syllabus reader that extracts deadlines and adds them to your calendar
- A course-schedule optimizer for next quarter
- A research-paper discovery and summarization agent
- A campus-event aggregator and recommender
- A coding agent specialized for a particular codebase, framework, or language
- A rigorous evaluation of an existing agent that surfaces its failure modes

Whatever you build, we expect rigor in how you evaluate it. Simply building a tool and showing that it works in one instance will not score highly. Strong projects define their evaluation early: what tasks the agent is expected to handle, where the data for those tasks comes from and how it was collected, what counts as success, and how success is measured (code-based graders, LLM-as-judge, human ratings, or a combination). Evaluate over a meaningful set of tasks, report results with appropriate baselines and comparisons, and analyze where and why the agent fails.

Teams & Mentors

Projects can be done in teams of 1–3 people. In our experience, teams of 3 lead to the best outcomes, so we encourage you to form a team of that size. Each team will be assigned a mentor from the teaching team, who will give feedback on all project-related work and generally be available throughout the quarter.

The proposal, midway report, and final report must use the ICLR submission format and follow the ICLR requirements, except where we specify otherwise below. Page limits exclude references.

AI-Use Disclosure Statement

Every stage of the project (proposal, midway report, and final report) must include an **AI-Use Disclosure Statement**, modeled on the ICLR policy on the use of large language models. ICLR requires a separate section describing the precise role of LLMs whenever they played a significant role in research ideation and/or writing; for this course, the statement is required at every stage even if your use was light or nonexistent (in which case, say so). Describe the precise role of any generative AI tools in that stage of the work: in ideation, in writing, and, since this is a systems course, in the code and experiments (e.g., writing or debugging code, generating or labeling data, analyzing results). As at ICLR, this section may appear in the appendix and does not count toward the page limit, and AI tools are not eligible for authorship. You take full responsibility for the contents submitted under your name, including content generated by AI tools that could be construed as plagiarism or fabrication. Using AI tools on the project is expected and consistent with the Honor Code & AI Tools policy under Logistics; the statement is about transparency. The statement is not graded on its own, but omitting it, or failing to disclose significant AI use, will cost points.

Code Submission

With the midpoint demo and the final report, you must also submit a link to a GitHub repository containing your project's code. The repository must include a README that walks a reader through running the project: how to install dependencies, configure any API keys or credentials, run the agent, and reproduce the evaluation reported in your paper. Write it so that a classmate who has never seen your code can get the system running. Your mentor will use the README when grading (reproducibility is scored separately from the final report), and it will also facilitate peer review of your project.

Milestones

Exact dates and times are listed in the Deadlines table, and the weight of each milestone is listed under Grading.

1. Project Proposal (Week 3)

A short paper (1–2 pages) that explains the problem your agent will address, why it matters, and how it fits within the scope of CS329Z. Briefly summarize existing work and compare it to your idea to justify how your project builds on it. The goal is to clearly define a direction; this is not a full literature review, but a focused justification of what you want to work on and why. The ideal is to keep the same topic for the proposal and the final project, but it is fine if your direction changes over time. The proposal is graded on its own merits regardless of what you end up building. Suggested structure:

1. **Problem Definition and Motivation:** What is the core problem your agent will address? Why is it meaningful, and to whom? Why is it a good fit for an agentic system rather than a simpler solution?
2. **Comparison with Existing Work:** What have past systems or papers tried in this area, and where do they fall short? This might be due to limitations in model capability, missing tools or data, poor reliability, or a lack of

how you would know if it works.

4. **References:** Entries should appear alphabetically and give at least full author name(s), year of publication, title, and outlet if applicable (e.g., journal or proceedings name).
5. **AI-Use Disclosure Statement:** See above.

2. Midpoint Demo & Midway Report (Week 7)

The midpoint demo is a recorded video of your system, submitted rather than presented in class; a working prototype is expected. Alongside the video, submit a link to your GitHub repository with a README for running the prototype (see Code Submission above). The midway report is a short, structured paper (3–4 pages) designed to help you establish your core system and evaluation framework. No prior-work discussion is needed. Grading is based mainly on the Environment & Data, Methods, and Summary of Progress sections. Required sections:

1. **Research Questions or Problem Specification:** A statement of the project's core research questions, or, for application-oriented projects, a specification of the problem the agent solves and for whom (one paragraph).
2. **Environment & Data:** A description of the environment, tools, and dataset(s) the agent will operate on, and the data you will use to evaluate it, including how that evaluation data is collected or constructed and what success criteria and metrics you will use.
3. **Methods:** A description of the agent architecture and approaches you are using, and a preliminary description of the approach that will be the focus of your investigation. At this stage, some aspects may not yet be worked out; preliminary descriptions are fine.
4. **Summary of Progress:** What you have done, what you still need to do, and any obstacles or concerns that might prevent your project from coming to fruition.
5. **References:** In the same format as for the proposal.
6. **AI-Use Disclosure Statement:** See above.

3. Final Report & Final System Demo (Finals Week)

The final system demo is presented live on Demo Day, Wednesday, December 9, 3:30–6:30 p.m. The final report is an 8-page paper in ICLR submission format, adhering to ICLR guidelines concerning references, layout, supplementary materials, and so forth. Submit it together with a link to your final GitHub repository, whose README must let a reader run the system and reproduce your results. The final submission is graded out of 25 points: 20 for the report itself and 5 for reproducibility (see below). Required components of the report:

1. **Introduction** (2 points)
2. **Related Work** (1 point)
3. **Environment & Data** (1 point)
4. **Methods** (5 points): The agent's architecture, components, and design decisions.
5. **Results** (10 points): Your evaluation of the system over a meaningful set of tasks, with baselines or comparisons, and an error analysis of its failure modes. A single working example is not an evaluation.
6. **Discussion / Conclusion** (1 point)

applications of your work. Here are some recommendations from ACL's ethics guidelines: "Ethical questions may arise when working with a variety of types of computational work with language, including (but not limited to) the collection and release of data, inference of information or judgments about individuals, real-world impact of the deployment of language technologies, and environmental consequences of large-scale computation."

8. **Authorship Statement:** At the end of your paper (after the Acknowledgments section in the template), include a brief authorship statement explaining how the individual authors contributed to the project. You are free to include whatever information you deem important to convey. For guidance, see the second page, right column, of this guidance for PNAS authors. We require this largely because we think it is a good policy in general. This statement is required even for singly-authored papers, because we want to know whether your project is a collaboration with people outside of the class. Only in extreme cases, and after discussion with the team, would we consider giving separate grades to team members based on this statement.
9. **References**
10. **AI-Use Disclosure Statement:** See above. Place it after the Authorship Statement.

The Safety & Ethical Considerations, Authorship Statement, References, and AI-Use Disclosure Statement sections are required but not separately scored; a missing section will cost points.

Reproducibility (5 points). The remaining 5 points of the final submission are for reproducibility. We will follow your README and run your code on a simple instance, such as a single task or example from your evaluation, to check that the system runs and behaves as described in the paper. We will not rerun your full evaluation, so make sure the quick-start path in your README works out of the box.